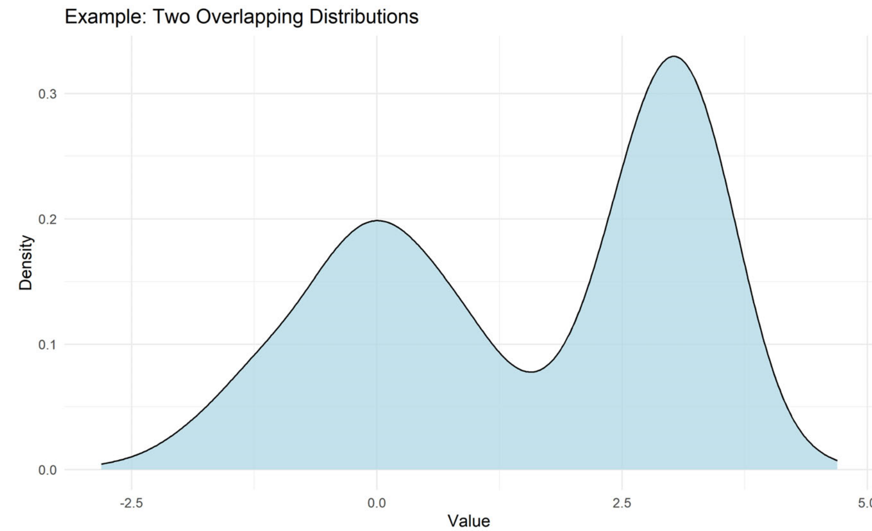
Descriptive Statistics: Numeric Edition

S. Mason Garrison

Characteristics of distributions

Describing distributions with numbers

- Hundreds of descriptive statistics exist
- Goal: Describe the data with a single number that represents an entire distribution
- Taxonomy of descriptive statistics (a.k.a. Key characteristics)
  - Shape
  - Center
  - Spread
  - Unusual observations

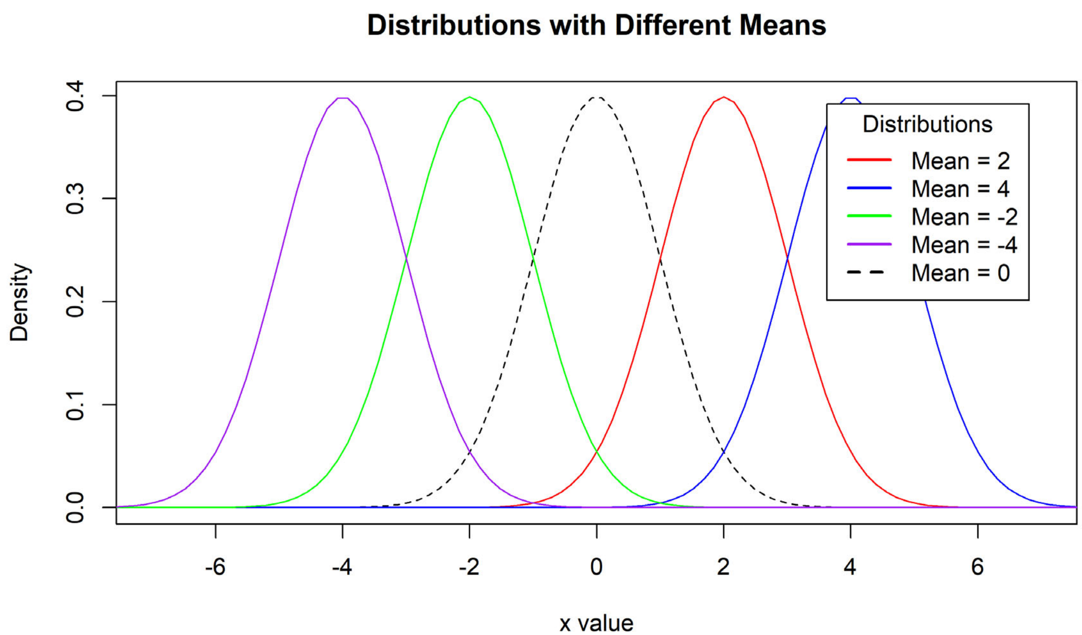


Taxonomy

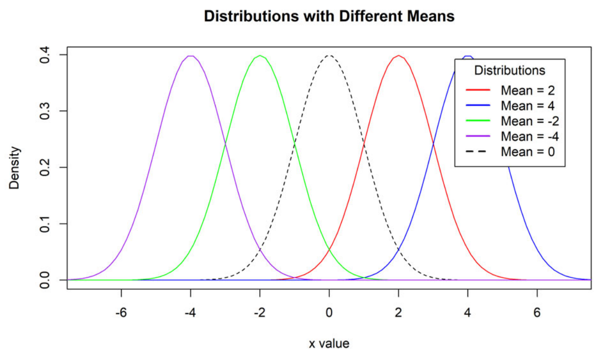
- shape:
  - skewness: right-skewed, left-skewed, symmetric (skew is to the side of the longer tail)
  - modality: unimodal, bimodal, multimodal, uniform
- center: mean (`mean`), median (`median`), mode (not always useful)
- spread: range (`range`), standard deviation (`sd`), inter-quartile range (`IQR`)
- unusual observations: outliers, extreme values

Center

- Center (or Central tendency or Location) aims to capture the center of the distribution.
- Display the normal distributions with various means



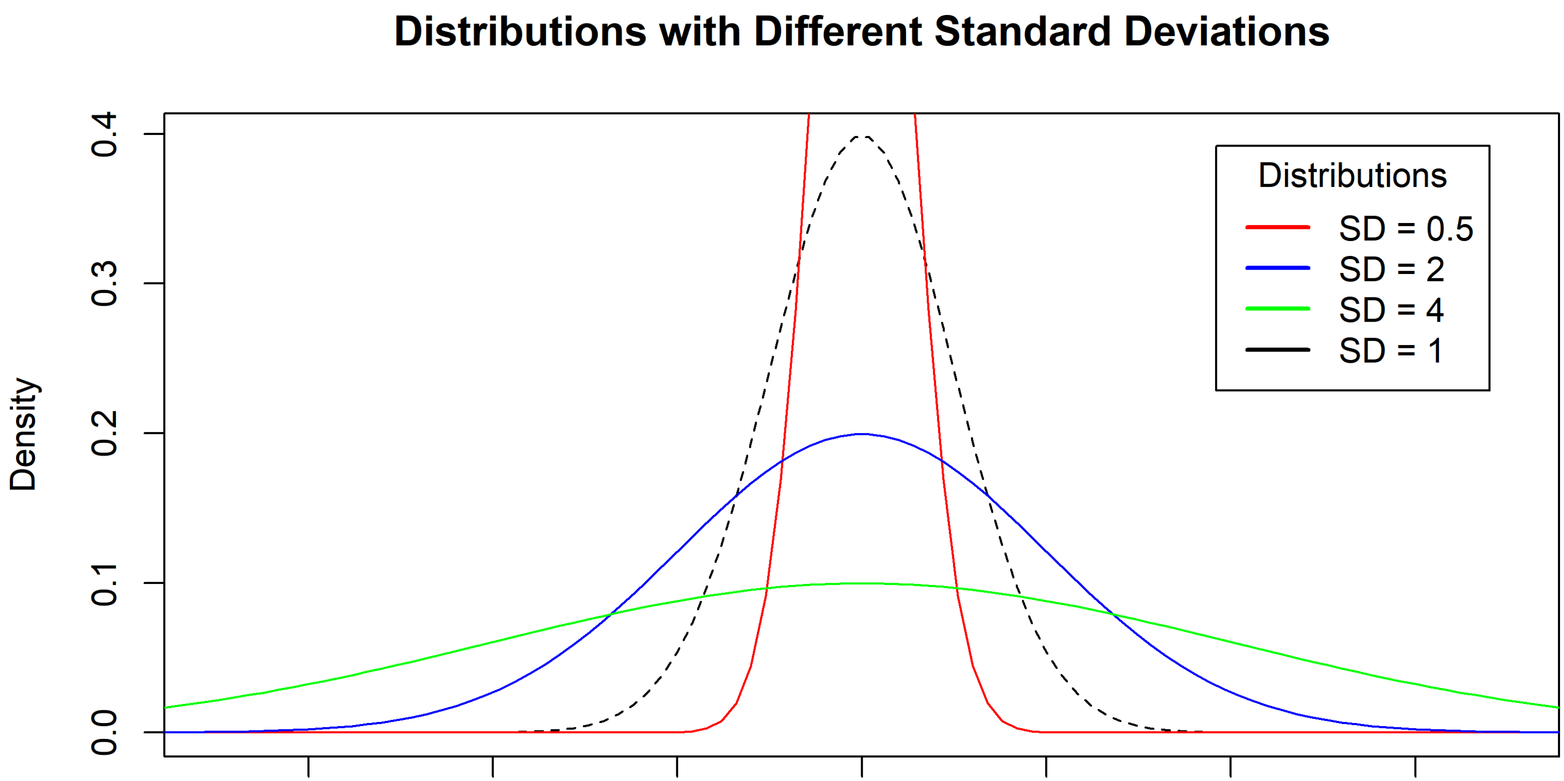
```r
x <- seq(-80, 80, length=1000)
hx <- dnorm(x)
colors <- c("red", "blue",
            "green", "purple", "black")


plot(x, hx, type="l", lty=2, xlab="x value",
         ylab="Density", main="Distributions with Different Means",xlim=c(-7, 7))
location<-c(2,4,-2,-4)
labels=paste0("Mean = ",location)
labels[length(labels)+1]="Mean = 0"

for (i in 1:length(location)){
    lines(x, dnorm(x,mean=location[i]), lwd=1, col=colors[i])
}

legend("topright", inset=.05, title="Distributions",
  labels, lwd=2, lty=c(1, 1, 1, 1, 2), col=colors)
```



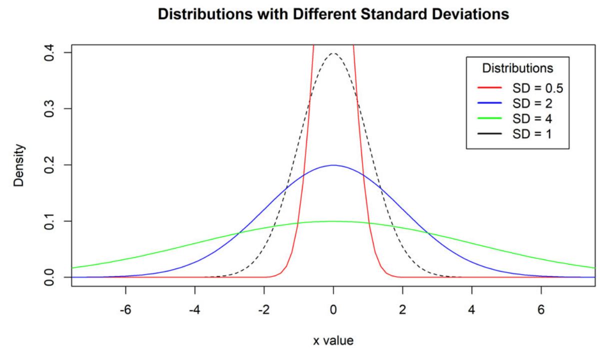
Spread

- Spread (a.k.a. Variability) describes how spread out the data are from that center
  - Low variance has a less wide distribution, with the bulk of the mass in the center
  - High variance has a very wide distribution, bulk of distribution is spread out

Normal distributions with various standard deviations



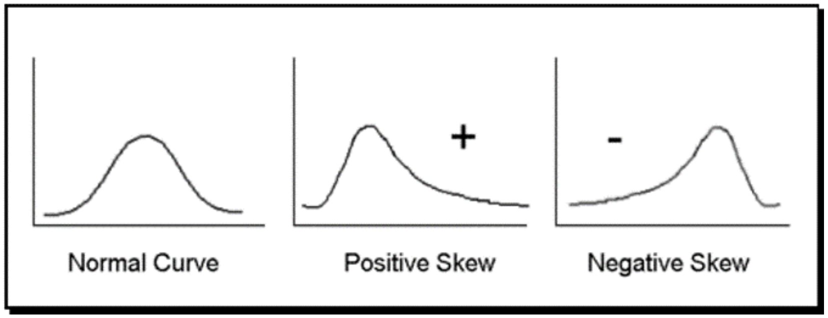
```r
plot(x, hx, type="l", lty=2, xlab="x value",
         ylab="Density", main="Distributions with Different Standard Deviations",xlim=c(-7, 7))
spread=c(.5,2,4)
labels=paste0("SD = ",spread)
labels[length(labels)+1]="SD = 1"
for (i in 1:length(spread)){
    lines(x, dnorm(x,sd=spread[i]), lwd=1, col=colors[i])
}
legend("topright", inset=.05, title="Distributions",
  labels, lwd=2, lty=c(1, 1, 1, 1, 2), col=colors[c(1:3,5)])
```



Skew / Asymetry

- Skewness is a measure of symmetry, or more precisely, the lack of symmetry.
- A distribution, or data set, is symmetric
  - if it looks the same to the left and right of the center point.
- Refers to the tail of the distribution

- Positively skewed (right skewed)
  - Bulk of the mass is on the left
- Negatively skewed (left skewed)
  - Bulk of the mass is on the right

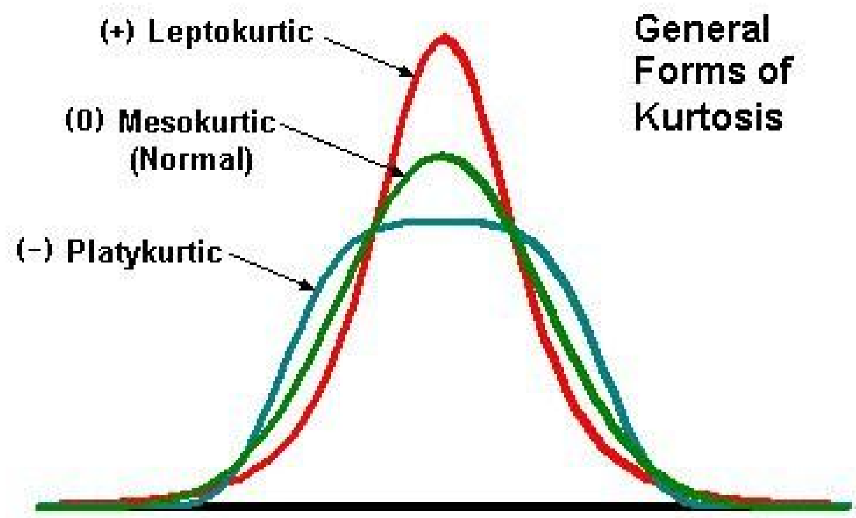


Kurtosis – peakedness

- Kurtosis is a measure of whether the data are
  - heavy-tailed or light-tailed relative to a normal distribution.
- Data sets with high kurtosis tend to have heavy tails, or outliers.
- Data sets with low kurtosis tend to have light tails, or lack of outliers.
  - A uniform (flat) distribution would be the extreme case.

Kurtosis – peakedness

- Low k
  - Leptokurtic
- Normal
  - Mesokurtic
- High k
  - Platykurtic



Specific Measures

- Measures of Central Tendency
  - Mean
  - Median
  - Mode

Central tendency: Mean

- Mean $(\mu; \bar{X})$
  - arithmetic average
  - $\bar{X}$ is used for samples
  - Mu (μ) is used for population

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Central tendency: Mean

- Properties
  - Is the balance point of the distribution (in terms of center of mass) $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$
  - Least squares property

- The sum of squared deviations about the mean is small

    - highly sensitive to outliers (extreme scores)
    - (weakness; it means that the mean is not so good as a measure of central tendency in highly skewed distributions)
    - Is not a robust statistic (low robust = sensitive to outliers; high robust = not sensitive to outliers)

Central tendency: Mean

- Very good with quantitative data (interval and ratio data,
  - especially bell shaped distributions)
  - Very popular statistic

Horse kick Data

```r
library(pscl)
data(prussian)

#horse kick fatalities by year
prussian$y
```

```
##   [1] 0 2 2 1 0 0 1 1 0 3 0 2 1 0 0 1 0 1 0 1 0 0 0 2 0 3 0 2 0 0
##  [31] 0 1 1 1 0 2 0 3 1 0 0 0 0 2 0 2 0 0 1 1 0 0 2 1 1 0 0 2 0 0
##  [61] 0 0 0 1 1 1 2 0 2 0 0 0 1 0 1 2 1 0 0 0 0 1 0 1 1 1 1 0 0 0
##  [91] 0 1 0 0 0 0 1 1 0 0 0 0 0 0 2 1 0 0 1 0 0 1 0 1 1 1 1 1 1 0
## [121] 0 0 1 0 2 0 0 1 2 0 1 1 3 1 1 1 0 3 0 0 1 0 1 0 0 0 1 0 1 1
## [151] 0 0 2 0 0 2 1 0 2 0 1 0 0 0 1 0 0 1 0 0 0 0 1 0 0 0 1 1 0 1
## [181] 0 0 0 0 0 2 1 1 1 0 2 1 1 0 1 2 0 1 0 0 0 0 1 1 0 1 0 2 0 2
## [211] 0 0 0 0 2 1 3 0 1 1 0 0 0 0 2 4 0 1 3 0 1 1 1 1 2 1 3 1 3 1
## [241] 1 1 2 1 1 3 0 4 0 1 0 3 2 1 0 2 1 1 0 0 0 1 0 0 0 0 0 1 0 1
## [271] 1 0 0 0 2 2 0 0 0 0
```

```r
# Mean
mean(variable)
```

```
## [1] 0.7
```

Central Tendency: Median

- Median (Md)
- Def: central score in a distribution
  - If *n is odd* then
    - Md = value of the $\frac{n+1}{2}$ item term.
  - If *n is even* then
    - Md = average of the $\frac{n+}{2}$ and $\frac{n+1}{2}$ item terms.

Central Tendency: Median

- Properties
  - Balance point of scores
  - Highly robust to outliers (less sensitive than the mean to outliers)
  - Sum of absolute deviations is smaller than any other constant (c) $\sum_{i=1}^{n}(|X - c|)$
- Often used for ordinal data

```
median(variable)
```

```
## [1] 0
```

Calculating by "hand"

- Sample Size?

```
(sample.size <- length(variable))
```

```
## [1] 280
```

- Even or Odd?
  - Test if number is divisible by 2.
  - If yes, then even
  - Else, is odd

```
sample.size %% 2 == 0
```

```
## [1] TRUE
```

- Sort our values

```
variable <- sort(variable)
```

- if odd, grab the midpoint value sample size / 2

```
variable[sample.size/2]
```

```
## [1] 0
```

- if even, grab the average of the midpoint values sample size / 2 and sample size + 1 / 2

```
(variable[sample.size/2] + variable[1+sample.size/2])/2
```

```
## [1] 0
```

Central Tendency: Mode

- Mode: most common score
- Local modes are the highest point with a subset of the distribution, there can be multiple ones

```
#Mode
mode(variable) # That's not mode!
```

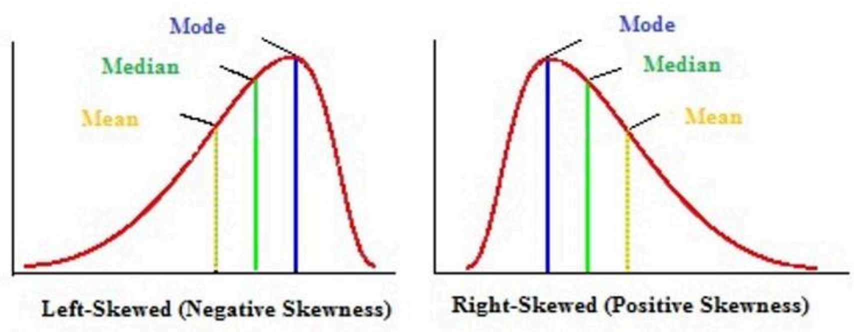
```
## [1] "numeric"
```

```
# Function to examine mode
Mode <- function(x) {
    ux <- unique(x)   #finds all un
    ux[which.max(tabulate(match(x,
}
Mode(variable)
```

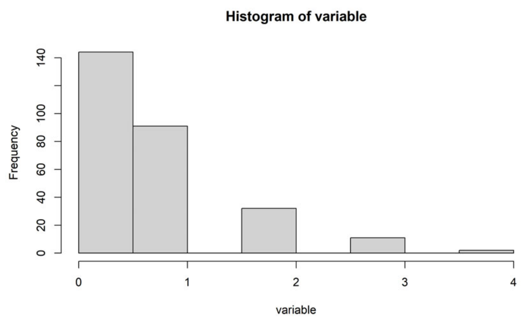
```
## [1] 0
```

Relationship between mean, median, and mode

- When we have a symmetric unimodal distribution
  - Mean=median=mode
- positively skewed
  - mode < median < mean
- negatively skewed
  - mean < median < mode



```r
hist(variable)
```




```r
library(Hmisc)
describe(variable)
```

```
## variable
##        n  missing distinct     Info     Mean    0.
##      280        0        5    0.828      0.7   0.
##
## Value          0     1     2     3     4
## Frequency    144    91    32    11     2
## Proportion 0.514 0.325 0.114 0.039 0.007
##
## For the frequency table, variable is rounded to t
```

Measures of Spread around the Median

- Range
  - maximum value - minimum value
  - $max(x_i) - min(x_i)$
  - Non-robust to outliers
- Quartiles
  - Lower quartile (Q1), 25th percentile
  - Second quartile (Q2), 50th percentile / median
  - Upper quartile (Q3), 75th percentile
- Interquartile Range (IQR)
  - Q3-Q1
  - Sometimes called h-spread; h = hinges

R Examples

```
# Range
range(variable)
```

```
## [1] 0 4
```

```
max(variable) -min(variable)
```

```
## [1] 4
```

```
summary(variable)# 5 Number Summary / Quartiles
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     0.0     0.0     0.0     0.7     1.0     4.0
```

Spread around the Mean

- Variance ($\sigma^{2}; s^{2}$)
  - Measure of spread around the mean
  - Goal of the measure to use every score

$\sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}$ $s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$

- Standard Deviation (σ;s)
  - $\sigma = \sqrt{\sigma^2}$
  - s = $\sqrt{s^2}$

Bessel's Correction

- s^2 is nearly the average squared deviation
- s^2 uses n-1 instead of N
  - Otherwise we get biased estimates
  - This adjustment is called Bessel's correction
- In our formula, we are using the sample mean (x) instead of the true mean (μ)
  - this results in underestimating each $x_i - \mu$ by $x - \mu$.

Properties

- Std is in raw score units
- Both the variance and standard deviation are highly NOT robust

R Examples

```
# Variance
var(variable)
```

```
## [1] 0.762724
```

```
# Standard Deviation
sqrt(var(variable))
```

```
## [1] 0.8733407
```

```
sd(variable)
```

```
## [1] 0.8733407
```

Wrapping Up...